

UI/UX Design Document

Project: ParkPulse – Smart Parking, Smarter Cities

Version: 2.0

Design Style: Modern Minimal Glassmorphism

Platform: Mobile First (React.js Responsive Web App)

Design Inspiration: Inspired by the modern parking app concept you shared from Dribbble. The interface will be **inspired by the visual language** (layout, spacing, hierarchy, and interactions) while using original assets, components, and implementation.

1. Design Vision

ParkPulse aims to create a **premium digital parking experience** where users can discover, reserve, navigate, and manage parking spaces effortlessly.

The interface should feel like a modern mobility application similar to premium navigation apps.

Design Goals

- Clean & Premium
 - Mobile First
 - Minimal User Flow
 - Fast Booking Experience
 - One-Hand Friendly
 - Highly Interactive
 - Real-Time Experience
 - Modern Card-Based Interface
-

2. Design Language

Theme

Modern Minimal

Design Style

- Soft White Background
 - Rounded Components
 - Floating Cards
 - Soft Shadows
 - Smooth Animations
 - Large Images
 - Interactive Maps
 - Clean Typography
 - Bottom Navigation
 - Gesture Friendly
-

3. Visual Identity

Brand Personality

- Smart
- Fast
- Reliable
- Premium
- Modern
- Friendly

User Experience

The application should make users feel like:

"Finding parking is now as easy as booking a cab."

4. Color Palette

Purpose	Color
Primary Green	#34C759
Primary Black	#111827
Secondary Gray	#6B7280
Background	#F6F7FB
Card Background	#FFFFFF

Border	#ECECEC
Available Slot	#34C759
Occupied Slot	#FF4D4F
Reserved Slot	#FFB020
Disabled Slot	#D1D5DB
Text Primary	#1F2937
Text Secondary	#6B7280

5. Typography

Primary Font

Poppins

Fallback

- Inter
- Sans-serif

Element	Size
App Title	32px
Heading	28px
Sub Heading	22px
Card Title	18px
Body	16px
Caption	14px
Small	12px

6. Icon System

Library

React Icons

Suggested Icons

- Home
 - Search
 - Parking
 - Car
 - QR Code
 - Calendar
 - Clock
 - Wallet
 - Notification
 - User
 - Map
 - Navigation
 - Settings
 - History
 - Filter
-

7. Application Navigation

Splash Screen



Login



Signup



Home

- Search
- Nearby Parking
- Parking Details
- Slot Selection
- Booking
- Active Parking
- Payment
- Booking History
- Notifications

8. Home Screen (Dashboard)

This is the first screen users see after login.

Header

- User Profile Image
- Greeting
- User Name
- Notification Icon
- Menu Button

Example

Welcome

Shristi 🖐️

Search Section

Large Search Bar

Placeholder

Search Parking Location...

Search Suggestions

- Shopping Mall
 - Airport
 - Hospital
 - Railway Station
 - Office
 - College
-

Category Chips

Horizontal Scroll

- All
 - Parking
 - Car
 - Bike
 - EV
 - Reserved
-

Google Maps Preview

Displays

- User Current Location
 - Nearby Parking Pins
 - Live Traffic
 - Navigation Button
-

Nearby Parking Cards

Each card contains

- Parking Image
- Parking Name
- Rating
- Address
- Distance
- Price per Hour
- Available Slots
- Book Now Button

Cards should scroll horizontally.

Bottom Navigation

- Home
 - Search
 - Reservations
 - Notifications
 - Profile
-

9. Parking Details Screen

Displays detailed information about the selected parking location.

Hero Section

Large Parking Image

Floating Favorite Button

Back Button

Share Button

Parking Information

Displays

- Parking Name
 - Address
 - Rating
 - Reviews
 - Price per Hour
 - Distance
 - Estimated Time
-

Amenities

Grid Layout

- CCTV
 - Security
 - EV Charging
 - Washroom
 - Car Wash
 - Wheelchair Access
-

Parking Statistics

Cards displaying

- Car Slots

- Bike Slots
 - EV Slots
 - Family Parking
-

Peak Hours Chart

Interactive graph showing busiest hours.

Bottom Action Button

Reserve Parking

10. Slot Selection Screen

One of the most important screens.

Displays a visual parking layout similar to a real parking lot.

Slot Colors

 Available

 Occupied

 Reserved

 Disabled

User Actions

Tap Available Slot

↓

Slot Selected

↓

Reservation Summary Opens

11. Reservation Screen

Displays

- Parking Name
- Slot Number
- Date
- Entry Time
- Estimated Duration
- Parking Charges

Buttons

- Confirm Booking
 - Cancel
-

12. QR Ticket Screen

After successful booking.

Displays

- QR Code
- Booking ID
- Slot Number
- Parking Name
- Entry Time
- Expiry Time

Buttons

- Navigate
 - Check-In
-

13. Active Parking Screen

Inspired by the third screen of the reference UI.

Displays

Parking Layout

Shows every parking slot.

Color Legend

 Available

 Occupied

 Reserved

Selected slot is highlighted.

Parking Timer

Large Circular Countdown

Displays

02:15 Remaining

Reminder Toggle

Notify me 15 minutes before expiry

Buttons

- Extend Parking
 - End Parking
-

14. Payment Screen

Displays

- Parking Duration
- Hourly Rate
- Total Cost
- Tax
- Discount (if any)

Payment Methods

- UPI
- Debit Card
- Credit Card
- Wallet

Button

Pay Now

15. Booking History

Displays previous bookings in card format.

Each card contains

- Parking Name
 - Date
 - Slot Number
 - Duration
 - Amount Paid
 - Payment Status
-

16. Notifications Screen

Displays

- Booking Confirmed
 - Reservation Expiring
 - Payment Successful
 - Parking Reminder
 - Offers & Discounts
-

17. Profile Screen

Displays

- Profile Picture
- Name

- Email
 - Phone
 - Saved Vehicles
 - Saved Payment Methods
 - Booking History
 - Settings
 - Logout
-

18. Admin Dashboard (Desktop)

Sidebar

- Dashboard
- Parking Lots
- Slot Management
- Reservations
- Payments
- Users
- Analytics
- Reports
- Settings

Dashboard Cards

- Total Slots
- Available Slots
- Occupied Slots
- Reserved Slots
- Daily Revenue
- Today's Bookings

Charts

- Occupancy Rate
 - Revenue Analysis
 - Peak Hours
 - Booking Trends
-

19. Components Library

Buttons

- Primary Button
 - Secondary Button
 - Icon Button
 - Floating Action Button
-

Cards

- Parking Card
 - Reservation Card
 - Booking Card
 - Payment Card
 - Statistics Card
-

Inputs

- Search Bar
 - Text Field
 - Dropdown
 - Date Picker
 - Time Picker
-

Status Chips

- Available
 - Reserved
 - Occupied
 - Open
 - Closed
-

Dialogs

- Booking Confirmation
 - Cancel Booking
 - Payment Success
 - Delete Confirmation
-

20. Micro Interactions

- Splash Screen Animation
 - Button Press Animation
 - Parking Card Hover
 - Slot Selection Animation
 - QR Code Generation Animation
 - Live Slot Status Update
 - Success Checkmark Animation
 - Smooth Screen Transitions
 - Skeleton Loading
 - Pull to Refresh
-

21. Responsive Design

Mobile

375px–430px

Primary Target Device

Tablet

768px–1024px

Desktop

1440px+

(Admin Dashboard)

22. Accessibility

- WCAG AA Color Contrast
- Keyboard Navigation
- Screen Reader Support
- Touch Targets (Minimum 44×44 px)
- Responsive Font Scaling
- Focus Indicators

23. UI Design Tokens

Property	Value
Border Radius	20px
Card Radius	24px
Button Radius	16px
Input Radius	16px
Shadow	0 10px 30px rgba(0,0,0,0.08)
Transition	300ms ease
Card Padding	20px
Screen Padding	24px
Grid Gap	20px

24. Recommended React Libraries

Purpose	Library
Routing	React Router DOM
State Management	Context API
Backend	Supabase
Maps	Google Maps JavaScript API
Icons	React Icons
Charts	Recharts
QR Code	react-qr-code
Notifications	React Toastify

Animations	Framer Motion
Forms	React Hook Form
Validation	Zod
HTTP Client	Supabase Client SDK

25. Recommended React Project Structure

src/

assets/

- | — images/
- | — icons/
- | — logos/

components/

- | — buttons/
- | — cards/
- | — navigation/
- | — search/
- | — parking/
- | — qr/
- | — timer/
- | — charts/
- | — dialogs/
- | — maps/

pages/

- | — Splash.jsx
- | — Login.jsx
- | — Signup.jsx
- | — Home.jsx
- | — Search.jsx
- | — ParkingDetails.jsx
- | — SlotSelection.jsx
- | — Booking.jsx
- | — Ticket.jsx
- | — ActiveParking.jsx
- | — Payment.jsx
- | — History.jsx
- | — Notifications.jsx

| — Profile.jsx
| — admin/

layouts/

hooks/

context/

services/

supabase/

utils/

constants/

styles/

App.jsx

main.jsx

26. Design Inspiration & Implementation Notes

The ParkPulse interface will be **inspired by the premium mobile parking concept** you shared, emphasizing a map-first experience, large parking cards, clean typography, rounded components, and intuitive parking slot visualization. The implementation will use original React components and styling while adopting the same design philosophy:

- Floating search and navigation controls
 - Card-based layout with soft shadows
 - Interactive parking map on the home screen
 - Rich parking detail page with images, statistics, and amenities
 - Color-coded parking slot grid for quick recognition
 - Circular active parking timer with reminders
 - Minimal bottom navigation for one-handed use
 - Smooth animations and responsive interactions using Framer Motion
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27. Final Technology Stack

Layer	Technology
Frontend	React.js (Vite)
Styling	Tailwind CSS
Backend	Supabase
Database	PostgreSQL (Supabase)
Authentication	Supabase Auth
Realtime	Supabase Realtime
Maps	Google Maps JavaScript API
Charts	Recharts
QR Code	react-qr-code
Animations	Framer Motion
Icons	React Icons
Forms	React Hook Form + Zod
Version Control	Git & GitHub
Deployment	Vercel

Conclusion

ParkPulse is designed to provide a **modern, premium, and intuitive smart parking experience**. The UI emphasizes speed, clarity, and ease of use through a clean card-based layout, real-time parking visualization, QR-based booking, and interactive maps. With React.js, Tailwind CSS, Supabase, and Vercel, the application will deliver a responsive, scalable, and production-ready user experience suitable for a capstone project and future real-world deployment.